

# Harsh Kumar

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[harshcurious.com](https://harshcurious.com) | [github.com/harshcurious](https://github.com/harshcurious)

## PROFESSIONAL SUMMARY

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**Decision Scientist & ML Engineer** with an **M.S. from IIT Kanpur** and 4+ years of experience building high-impact models for **massive-scale** e-commerce and fintech environments. Expert in developing **real-time risk engines** and **forecasting systems** managing **\$100M+ in annual volume**. Proven track record in transitioning complex research into **low-latency (200-400ms)** production code using **Databricks, PySpark, and Airflow**. Specialized in end-to-end ML lifecycles and automated **CI/CD** orchestration.

## SKILLS

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**ML & AI:** XGBoost, LightGBM, Transformers, PyTorch, **Time series forecasting (fbprophet)**, Reinforcement Learning, Optuna, SHAP, Scikit-learn, LangChain, Langgraph, Pydantic AI.

**Data & Infrastructure:** **Databricks, PySpark, Airflow, GitLab CI/CD**, BentoML, MLFlow, Docker, Kubernetes, DynamoDB, Postgres, Snowflake, AWS SageMaker.

**Languages:** Python (Expert), C, Go, SQL.

## WORK EXPERIENCE

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### DECISION SCIENTIST — ONE

Bengaluru, India — 2024 – 2026

- **Real-Time Transaction Systems:** Architected a hybrid pipeline utilizing **DynamoDB** for sub-second retrieval of features, optimizing **latency** for real-time risk assessment at scale.
- **Scalable ML Infrastructure:** Standardized organization-wide ML deployment using **GitLab CI/CD** and **Airflow** DAGs within **Databricks**, automating re-training and monitoring (PSI/CSI).
- **High-Scale Feature Engineering:** Built a **PySpark** library with 2,000+ signals, handling massive datasets to improve default risk estimation (XGBoost).
- **Credit Risk Modeling:** Designed models managing 2M+ transactions (\$100M+ volume), maintaining default rates below industry standards and maximizing product profitability.

### DATA SCIENTIST — ZESTMONEY

Bengaluru, KA, IN — 2022 – 2023

- **Demand Forecasting:** Designed a statistical time-series forecasting engine (**fbprophet**) to optimize workforce management, reducing peak-hour disconnect rates from 50% to <20%.
- **Model Serving & Latency:** Deployed real-time classification APIs via **BentoML** with **200-400ms latency**, aggregating 2,000+ features from 8 disparate data sources.
- **Propensity & Growth:** Built a suite of **Spark-based XGBoost** models for product cross-sell, achieving up to **30% uplift** in conversion across evolving product lines.
- **ETL Optimization:** Improved pipeline efficiency by **40%** by migrating legacy Pandas workflows to **Polars** for high-speed feature extraction.

## EDUCATION

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**M.S. in Mathematics and Scientific Computing**

GPA: 9.1

**IIT Kanpur**

Jul 2016 – Jun 2017

**B.S. in Mathematics and Scientific Computing**

GPA: 8.4

**IIT Kanpur**

Jul 2012 – Jun 2016